130. *(a)* Given, diameter of a sphere, $d = 6$ cm

$\therefore$ Radius of a sphere, $r = \frac{d}{2} = \frac{6}{2} = 3$ cm

Let the radius of wire be R cm.

Also, given the length of wire, $H = 36$ m $= 3600$ cm

According to the question,

Volume of sphere = Volume of wire

$$\Rightarrow \frac{4}{3}\pi r^3 = \pi R^2 H$$

$$\Rightarrow \frac{4}{3}\times(3)^3 = R^2 \times 3600$$

$$\Rightarrow R^2 = \frac{4\times 3^2}{3600} = \frac{(6)^2}{(60)^2}$$

$$\therefore R = \frac{6}{60} = \frac{1}{10} = 0.1 \text{ cm}$$

131. *(b)* Given, side of a cube = 2 cm

$\therefore$ Maximum distance between two points of a cube

= Length of diagonal

$= \sqrt{3} \times$ Side $= 2\sqrt{3}$ cm

132. *(d)* Let the slant height of cone be l cm.

Given, $h = 24$ cm and $r = 7$ cm

$$\therefore l^2 = h^2 + r^2$$

$$\Rightarrow l^2 = 24^2 + 7^2 = 576 + 49$$

$$\Rightarrow l^2 = 625 \Rightarrow l = 25 \text{ cm}$$

Total surface area = Curved surface area of cone + curved surface area of hemisphere

$$= \pi r l + 2\pi r^2 = \pi r(l + 2r)$$

$$= \pi \times 7(25 + 2\times 7) = 7\pi[25 + 14]$$

$$= 7\pi \times 39 = 273\pi \text{ cm}^2$$

133. *(a)* Let the radius of the sphere and side of cube be r and a respectively.

$\therefore$ Total surface area of sphere

$(S_1) = 4\pi r^2$ sq units

and total surface area of cube

$(S_2) = 6a^2$ sq units

Now, according to the question,

$$S_1 = S_2 \Rightarrow 4\pi r^2 = 6a^2 \Rightarrow \frac{r^2}{a^2} = \frac{6}{4\pi} \quad \text{...(i)}$$

$\therefore$ Volume of sphere $(V_1) = \frac{4}{3}\pi r^3$ cu units

and volume of cube $(V_2) = a^3$ cu units

Now,

$$\left(\frac{V_1}{V_2}\right)^2 = \left(\frac{\frac{4}{3}\pi r^3}{a^3}\right)^2 = \frac{16}{9}\pi^2\left(\frac{r^2}{a^2}\right)^3$$

$$= \frac{16}{9}\pi^2\left(\frac{6}{4\pi}\right)^3 \quad \text{[from Eq. (i)]}$$

$$= \frac{16}{9}\pi^2 \times \frac{6\times 6\times 6}{4\times 4\times 4\times \pi^3}$$

$$= \frac{6}{\pi} = 6 : \pi$$

Hence, the ratio of square of their volume is $6 : \pi$.

134. *(a)* Let radius of sphere and cone be r.

$\therefore$ Volume of sphere $V_1 = \frac{4}{3}\pi r^3$

and volume of cone $V_2 = \frac{1}{3}\pi r^2 h$

Now, according to the question,

$$V_1 = 2V_2 \Rightarrow \frac{4}{3}\pi r^3 = 2\times\frac{1}{3}\pi r^2 h$$

$$\Rightarrow 2r^3 = r^2 h \Rightarrow 2r = h \Rightarrow \frac{h}{r} = \frac{2}{1}$$

$$\therefore h : r = 2 : 1$$

135. *(a)* Let the radius of sphere be r.

$\therefore$ Volume of sphere $= \frac{4}{3}\pi r^3$

According to the question,

If the radius of a sphere is increased by 10%.

Then, new radius $r' = r + r\times 10\%$

$$= r + \frac{r}{10} = \frac{11r}{10}$$

$\therefore$ New volume of sphere $= \frac{4}{3}\pi r'^3$

$$= \frac{4\pi}{3}\left(\frac{11r}{10}\right)^3 = \frac{4}{3}\pi \times \frac{1331r^3}{1000}$$

Increased volume

$$= \frac{4}{3}\pi\times\frac{1331}{1000}r^3 - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi r^3 \times \frac{331}{1000}$$

Increased percentage

$$= \frac{\frac{4}{3}\pi r^3 \times \frac{331}{1000}}{\frac{4}{3}\pi r^3}\times 100\% = 33.1\%$$

136. *(b)* Since, three metallic spheres are melted to form a single sphere.

So, the sum of volume of three spherical solid sphere.

$$= \frac{4}{3}\pi(6)^3 + \frac{4}{3}\pi(8)^3 + \frac{4}{3}\pi(10)^3$$

$$= \frac{4}{3}\pi[6^3 + 8^3 + 10^3]$$

Let r be the radius of new sphere

$$\therefore \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(6^3 + 8^3 + 10^3)$$

$$\Rightarrow r^3 = 6^3 + 8^3 + 10^3$$

$$\Rightarrow r^3 = 216 + 512 + 1000$$

$$\Rightarrow r^3 = 1728$$

$$\Rightarrow r^3 = 12\times 12\times 12$$

$$\Rightarrow r = 12 \text{ cm}$$

$\therefore$ Diameter of the new sphere

$= 2r = 2\times 12 = 24$ cm

137. *(b)* Let the radius and height of the cone be r and h, respectively.

$\therefore$ Initial volume of cone

$$(V) = \frac{1}{3}\pi r^2 h \quad \text{...(i)}$$

new height, $H = h + h\times 200\%$

$$= h + \frac{200h}{100} = 3h$$

and new radius, $R = r - r\times 50\%$

$$= r - \frac{50r}{100} = \frac{r}{2}$$

$\therefore$ New volume of cone $(V_2) = \frac{1}{3}\pi R^2 H$

$$= \frac{1}{3}\pi\left(\frac{r}{2}\right)^2 \times 3h = \frac{1}{3}\pi\times\frac{r^2}{4}\times 3h$$

$$= \frac{1}{3}\pi r^2 h \times \frac{3}{4} = \frac{3V}{4} \quad \text{[from Eq. (i)]}$$

Decrease in volume $= V - \frac{3V}{4} = \frac{V}{4}$

$\therefore$ Decrease percentage in volume

$$= \frac{\text{Decrease in Volume}}{\text{Initial Volume}}\times 100\%$$

$$= \frac{V/4}{V}\times 100\% = \frac{V}{4V}\times 100\% = 25\%$$

138. *(c)* Given a rectangular paper of 44 cm long and 6 cm wide is rolled to form a cylinder of height equal to width of the paper.

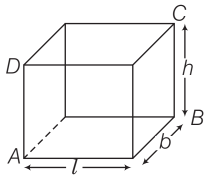
$\therefore$ Circumference of the base of cylinder $= 44$

i.e. $2\pi r = 44 \Rightarrow r = \frac{44}{2\pi} = \frac{44\times 7}{2\times 22}$

$\therefore r = 7$ cm

Hence, the radius of the base of the cylinder is 7 cm.

139. *(c)* Let the dimensions of cuboid be l, b and h respectively.



For first face

$$l^2 + b^2 = 13^2 \quad [\because l^2 + b^2 = d^2]$$

$$\Rightarrow l^2 + b^2 = 169 \quad \text{...(i)}$$

For second face

$$b^2 + h^2 = (\sqrt{281})^2$$

$$\Rightarrow b^2 + h^2 = 281 \quad \text{...(ii)}$$

For third face

$$h^2 + l^2 = 20^2 \Rightarrow h^2 + l^2 = 400 \quad \text{...(iii)}$$

On adding Eqs. (i), (ii) and (iii), we get

$$2(l^2 + b^2 + h^2) = 850$$

$$\Rightarrow l^2 + b^2 + h^2 = 425 \quad \text{...(iv)}$$

Subtracting Eq. (i) form Eq. (iv), we get

$h^2 = 256 \Rightarrow h = 16$ units

similarly, $b^2 = 25 \Rightarrow b = 5$ units

and $l^2 = 144 \Rightarrow l = 12$ units

$\therefore$ Total surface area of cuboid

$= 2(lb + bh + lh)$

$= 2(12 \times 5 + 5 \times 16 + 16 \times 12)$
$= 2(60 + 80 + 192)$
$= 2 \times 332 = 664$ sq units

Hence, the total surface area of cuboid is 664 sq units.

140. *(c)* Given, radius of vessel, $(r) = 4$ cm and radius of sphere $(R) = 3$ cm

Let height of vessel be h cm, then volume of vessel$= \pi r^2 h = \pi(4^2)h = 16\pi h$

and volume of sphere $= \frac{4}{3}\pi R^3 = 36\pi$

Let H be the rise in water level

$\therefore$ Volume of water displaced
= volume of sphere

$\Rightarrow 16\pi H = 36\pi \Rightarrow H = \frac{36}{16} = 2.25$ cm

141. *(c)* Volume of paint required

$= 2 \times (3 \times 2 \times 0.0001) + 2 \times (2 \times 1.75 \times 0.0001) + 2 \times (3 \times 1.75 \times 0.0001)$

$= 0.0012 + 0.0007 + 0.00105 = 0.00295 \text{ m}^3$

Volume of cubical box $= 10^3$.
$= 1000 \text{ cm}^3 = 0.001 \text{ m}^3$

$\therefore$ Boxes required $= \frac{0.00295}{0.001} \approx 3$.

142. *(d)* Let the sides of a cube be l cm.

Then, $6l^2 = 13254$

$\Rightarrow l^2 = \frac{13254}{6} = 2209 \Rightarrow l = 47$

$\therefore$ Length of diagonal $= l\sqrt{3} = 47\sqrt{3}$ cm

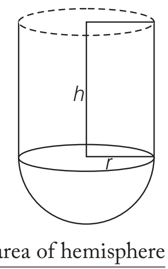
143. *(b)* Let the height of cylindrical portion be h.

Then, $\frac{2}{3}\pi r^3 + \pi r^2 h = 3312\pi$

$\Rightarrow \pi r^2\left[\frac{2}{3}r + h\right] = 3312\pi$

$\Rightarrow 12 \times 12\left[\frac{2}{3} \times 12 + h\right] = 3312$

$\Rightarrow 8 + h = 23 \Rightarrow h = 15$m



$\because \frac{\text{Surface area of hemisphere}}{\text{Surface area of cylinder}} = \frac{2\pi r^2}{2\pi rh}$

$= \frac{r}{h} = \frac{12}{15} = \frac{4}{5}$

$\therefore r : h = 4 : 5$

144. *(a)* Let the length of water tank be lm. volume of tank $= l \times l \times l = l^3$

volume of water tank after taking out water

$= l \times l \times (l - 2) = l^2(l - 2)$

$\therefore l^3 - 128 = l^2(l - 2)$

$\Rightarrow 2l^2 = 128 \Rightarrow l^2 = 64 \Rightarrow l = 8$ m

So, volume of cubical tank
$= 8^3 = 512 \text{ m}^3$.

145. *(b)* Let the sides of the cuboid be l, b and h respectively.

Volume of cuboid,

$V = l \times b \times h$ ...(i)

Given, $x = lb$, $y = bh$ and $z = lh$

$\therefore xyz = lb \times bh \times lh$
$= (lbh)^2 = V^2$ [using Eq. (i)]

146. *(a)* On each face, we can draw four right angled triangles by choosing one vertex to serve as the right angled and two adjacent vertices. Since there are six faces, you can form $6 \times 4 = 24$ triangles.

147. *(a)* Let the area of each rectangle be A.

Then, breadth of R_1, R_2 and R_3 are $\frac{A}{x_1}$, $\frac{A}{x_2}$ and $\frac{A}{x_3}$ respectively

Cylinder is formed by joining parallel side of breadth.

Here, length becomes height

and $2\pi r =$ breadth $= \frac{A}{x} \Rightarrow r = \frac{A}{2\pi x}$

$\therefore V = \pi r^2 h = \pi\frac{A^2}{4\pi^2 x^2} \times x = \frac{A^2}{4\pi x}$

Now, V_1, V_2 and V_3 are $\frac{A^2}{4\pi x_1}$, $\frac{A^2}{4\pi x_2}$

and $\frac{A^2}{4\pi x_3}$ respectively

$\because x_1 < x_2 < x_3 \Rightarrow \frac{1}{x_1} > \frac{1}{x_2} > \frac{1}{x_3}$

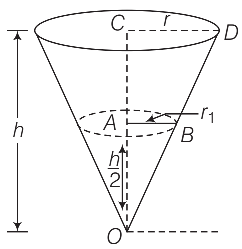
$\Rightarrow \frac{A^2}{4\pi x_1} > \frac{A^2}{4\pi x_2} > \frac{A^2}{4\pi x_3}$

$\therefore V_3 < V_2 < V_1$

148. *(a)* Since, ΔOAB and ΔOCD are similar

$\therefore \frac{r}{r_1} = \frac{h}{\frac{h}{2}} = \frac{2}{1}$

We have, $V = \frac{1}{3}\pi r^2 h$



$\Rightarrow r^2 = \frac{3V}{\pi h} \Rightarrow r_1 = \frac{r}{2} \Rightarrow r_1^2 = \frac{r^2}{4}$

Hence, volume of water $= \frac{1}{3}\pi r_1^2 \frac{h}{2}$

$= \frac{1}{6}\pi h\frac{r^2}{4} = \frac{1}{24}\pi h\frac{3V}{\pi h} = \frac{3V}{24} = \frac{V}{8}$

149. *(b)* Let radius be r and height be h of the cylinder.

Then, $30 \cdot \pi r^2 h = \frac{1}{3}\pi r^2 h \times n$

$\Rightarrow n = 30 \times 3 = 90$ cones

Statement I is correct. Statement II is also correct. But statement II is not correct explanation of statement of I.

150. *(c)* At 30 min, the container was full

$\therefore$ At 5 min before i.e. in 25 min, the container was half filled and at 20 min, container was one-fourth full.

151. *(c)* I. Area of curved surface $= 2\pi rh$

$\because 2\pi r(2h) = 2 \cdot 2\pi rh$

$\therefore$ Curved surface area is also doubled.

It is correct.

II. Total surface area of hemisphere $= 3\pi r^2$

$\because 3\pi(2r)^2 = 3\pi 4r^2 = 4 \cdot 3 \cdot \pi r^2$
= 4 times the area

It is also correct.

152. *(c)* Let n spherical ball be made.

Then, $4\pi r^2 = \pi^{\frac{1}{3}}$

$\Rightarrow r^2 = \frac{\pi^{\frac{1}{3}}}{4\pi} = \frac{1}{4}\pi^{-\frac{2}{3}} \Rightarrow r^2 = \left(\frac{1}{2}\pi^{-\frac{1}{3}}\right)^2$

$\Rightarrow r = \frac{1}{2}\pi^{-\frac{1}{3}}$

Given, $n \cdot \frac{4}{3}\pi\left(\frac{1}{2}\pi^{-\frac{1}{3}}\right)^3 = 1 \times 1 \times 1$

$\Rightarrow n \cdot \frac{4}{3}\pi \times \frac{1}{8} \times \frac{1}{\pi} = 1 \Rightarrow \frac{n}{6} = 1$

$\therefore n = 6$

153. *(d)* Let n cylindrical boxes be packed.

$\Rightarrow n \times \pi \times 5 \times 5 \times 10$
$= \pi \times 15 \times 15 \times 100$

$\therefore n = 90$ boxes

154. *(c)* We have, area of cross-section of pipe
$= 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$

and $4 \times 10^{-4} \times 40 \text{ m}^3$ water flows in 1 s.

Let cistern fill in t s.

$\Rightarrow 4 \times 10^{-4} \times 40 \times t = 10 \times 8 \times 6$

$\Rightarrow t = 30000 \text{ s} = \frac{30000}{60 \times 60}$ h

$= \frac{50}{6}$ h $= 8$ h and 20 min

28

STATISTICS

Usually (10-13) questions have been asked from this chapter. This chapter is very important from examination point of view and generally questions are asked from the graphical representation of data, measures of central tendency.

Statistics is the branch of Mathematics which deals with the collection, analysis and interpretation of numerical data. In this chapter, we shall study measures of central tendency i.e. mean, median and mode of ungrouped data and grouped data. Concept of cumulative frequency, the cumulative frequency distribution how to draw cumulative frequency curves (ogive) and graphical representation of data will also be discussed.

Collection of Data

Collection of data is the first step in statistics towards achieving the goal or conclusion.

On the basis of collection, data are of two types

1. **Primary data** The data collected actually in the process of investigation by the investigator is called primary data. It is original and first hand information.
2. **Secondary data** The data collected by someone and used by any other person known as secondary data.

Presentation of Data

Raw or Ungrouped data When the data presented is random and is not prepared according to some order, it is known as raw or ungrouped data. It does not give us a clear picture of the class.

Grouped data When the data is arranged in any manner like ascending or descending order etc., it is called grouped data. It can also be presented in the form of a table called frequency distribution table.

Class Intervals Class intervals are the groups in which all the observations are divided. Each class is bounded by two figures (numbers) which are called class limits. The figure on the left side of a class, is called its lower limit and that on the right side of a class, is called its upper limit.

Class mark It is the mid-point of the class interval.

i.e. $\text{Class mark} = \dfrac{\text{Lower class limit} + \text{Upper class limit}}{2}$

Range or a class size Difference between the upper limit and the lower limit of a class is called its class size.

Range = Upper limit – Lower limit

e.g. Range of the observations 4, 7, 8, 10, 12
$= 12 - 4 = 8$

Frequency of an observation The number of times an observation occurs is called its frequency.

Frequency distribution

The tabular arrangement of data, showing the frequency of each observation is called a frequency distribution. It is a method of presenting the data in a summarized form. Frequency distribution is also known as frequency table.

There are two types of frequency distribution which are as follows:

1. Discrete frequency distribution

A frequency distribution is called a discrete frequency distribution, if data are presented in a way such that exact measurements of the units are clearly shown.

Marks	Number of students (frequency)
40	7
60	3
80	3
100	2
Total	15

2. Continuous frequency distribution

Exclusive form A frequency distribution in which upper limit of each class is excluded and lower limit is included, is called an exclusive form.
e.g. In the class 0-10 of marks obtained by students, a student who has obtained 10 marks is not included in this class. It will be counted in the next class i.e 10-20.

Inclusive method In this method,the classes are so formed that the upper limit of a class is included in that class. The following example illustrates the method. In class 1000-1099, we include workers having wages between ₹ 1000 and ₹ 1099. If the income of a worker is exactly ₹ 1100, then it will be included in the next class 1100-1199.

Exclusive method		Inclusive method	
Wages (in ₹)	**Number of workers**	**Wages (in ₹)**	**Number of works**
1000-1100	125	1000-1099	125
1100-1200	150	1100-1199	150
1200-1300	200	1200-1299	200
1300-1400	250	1300-1399	250
1400-1500	175	1400-1499	175
1500-1600	100	1500-1599	100
Total	1000	**Total**	1000

It is clear from the above example that both the inclusive and exclusive methods give us the same class frequency although the class intervals are aparently different in the two cases.

In the above example on inclusive method, the difference between the lower limit of a class and upper limit of the preceding class is 1.

Therefore, we subtract $\frac{1}{2}$ from the lower limit and add $\frac{1}{2}$ to upper limit of each class to make it continuous.

The adjusted classes would be as follows:

Wages (in ₹)	Number of workers
999.5-1099.5	125
1099.5-1199.5	150
1199.5-1299.5	200
1299.5-1399.5	250
1399.5-1499.5	175
1499.5-1599.5	100

Cumulative frequency

If the frequency of first class interval is added to the frequency of second class and this sum is added to third class and so on, then frequencies so obtained are known as cumulative frequency.

EXAMPLE 1. Consider the table given below :

Marks	Number of students (frequency)	Cumulative frequency
0-10	13	13
10-20	7	20
20-30	5	25
30-40	4	29
40-50	1	30
50-60	7	37
60-70	3	40
70-80	4	44
80-90	5	49
90-100	1	50
Total	50	

Then, find the value of the following
(i) frequency of class 10-20. (ii) class size.
(iii) mid value of 60-70. (iv) total frequencies.

a. 10, 65, 50, 60 b. 7, 10, 65, 60
c. 50, 65, 10, 10 d. 7, 10, 65, 50

Sol. d. (i) Here, frequency of class 10-20 is 7.

(ii) Class size = Upper limit − Lower limit $= 30 - 20 = 10$

(iii) Mid value $= \frac{\text{Upper limit} + \text{Lower limit}}{2} = \frac{60 + 70}{2} = 65$

(iv) Total frequencies = 50

EXAMPLE 2. The class mark of the interval 12.5-17.5 is

a. 5 b. 12.5 c. 15 d. 17.5

Sol. c. Class mark $= \frac{\text{Lower limit} + \text{Upper limit}}{2}$

$$= \frac{12.5 + 17.5}{2} = \frac{30}{2} = 15$$

EXAMPLE 3. The class marks of a distribution are 54, 64, 74, 84, 94 and 104. Then, the class size is

a. 5 b. 10 c. 54 d. 104

Sol. b. Since, class size is the difference between the class marks of two adjacent classes.

$\therefore$ Class size $= 64 - 54 = 10$

MEASURES OF CENTRAL TENDENCY

An average or central value of a statistical series is the value of the variable which describes the charateristic of the entire distribution. The following are the five measure of central tendency

(i) Arithmetic mean or mean (ii) Geometric mean
(iii) Harmonic mean (iv) Median
(v) Mode

Out of these measures of central tendency. Arithmetic mean, median and mode are sometimes knows as measures of location.

Arithmetic Mean

The mean (or average) of number of observations is the sum of the values of all the observations divided by the total number of observations.

Mean = Sum of observations/Number of observation

Properties of Arithmetic Mean

1. If every observation is increased by a constant, then the mean of the observations, so obtained also increases by the same constant.
2. If every observation is decreased by a constant, then the mean of the observation, so obtained also decreases by the same constant.
3. If each observation is multiplied by a constant, then the mean of the resulting observations can be obtained by multiplying the mean by the same constant.
4. If each observation is divided by a constant, then the mean of the resulting observation can be obtained by dividing the mean by the same constant.
5. The average of natural numbers from 1 to n is $\frac{(n+1)}{2}$.
6. The average of odd numbers from 1 to n is $\left[\frac{\text{Last odd number}+1}{2}\right]$ and the average of even numbers from 1 to n is $\left[\frac{\text{Last even number}+2}{2}\right]$.

1. **Arithmetic mean of ungrouped or individual observations** If $x_1, x_2, x_3, \ldots, x_n$ are n observations, then

(i) Mean $(\bar{x}) = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$

It is also called direct method.

(ii) Mean $(\bar{x}) = A + \frac{1}{n}\sum_{i=1}^{n} d_i$

where, A = Assumed mean and $d_i = x_i - A$.

It is also called shortcut method.

EXAMPLE 4. If the heights of 5 persons are 144 cm, 152 cm, 150 cm, 158 cm and 155 cm, respectively. Find the mean height.

a. 152.5 cm b. 150 cm c. 149.8 cm d. 151.8 cm

Sol. d. Mean height $= \frac{\text{Sum of the heights}}{\text{Number of persons}}$

$= \frac{144+152+150+158+155}{5} = \frac{759}{5} = 151.8$ cm

EXAMPLE 5. If the average of 6, 8, 5, 7, x and 4 is 7. Then, the value of x is

a. 10 b. 11 c. 12 d. 15

Sol. c. $\because$ Sum of observation $= 6+8+5+7+x+4 = 30+x$

and number of observation $= 6$

$\therefore$ Average $= \frac{30+x}{6} \Rightarrow 7 = \frac{30+x}{6}$

$\Rightarrow 42 = 30 + x \Rightarrow x = 42 - 30 = 12$

Hence, the value of x is 12.

EXAMPLE 6. The average of 27 observations is 35. If 5 is added to each observation, what will be the new mean?

a. 10 b. 20 c. 30 d. 40

Sol. d. Given, $\bar{x} = 35$ and $n = 27$

Sum of observation $= n\bar{x} = 27 \times 35 = 945$

and new total of observation $= 945 + 27 \times 5 = 1080$

$\therefore$ New mean $= \frac{1080}{27} = 40$

Shortcut Method

New mean = Previous mean + Number added to each term $= 35 + 5 = 40$

EXAMPLE 7. The mean of 53 observations is 18. If each observation is multiplied by 3. What will be the new mean?

a. 18 b. 36 c. 53 d. 54

Sol. d. Given, $\bar{x} = 18$ and $n = 53$

So, sum of observation $= n\bar{x} = 53 \times 18 = 954$

New total of observation $= 954 \times 3 = 2842$

$\therefore$ New mean $= \frac{2862}{53} = 54$